

# Doohickey blueprint

Sirusdog

July 30, 2026

## 1 Basic installation

### 1.1 Initial setup

1. Flash an SD card with Rasbian Lite using the raspberry pi imager. Keep in mind that this code should work on other hardware/os versions, but setup will likely differ.
2. Plug in the RPI, connect it to a display via HDMI and give it power via the USB-C port.
3. I'd recommend leaving the username as pi and setting a sudo password you'll remember.
4. Connect the RPI to the internet, either via it's ethernet port or WIFI.

### 1.2 Software installs and interface setup

1. Run `sudo apt install git`
2. Run `git clone https://github.com/Sirusdog/Thermal-camera-project.git`
3. Run `cd Thermal-Camera-Project`
4. Run `sudo raspi-config`
  - (a) Select *3 Interface Options*
  - (b) Enable *I4 SPI*
  - (c) Enable *I5 I2C*
  - (d) Select *I6 Serial port*
    - i. Select *No* for "Would you like a login shell to be accessible over serial?"
    - ii. Select *Yes* for "Would you like the serial port hardware to be enabled?"
  - (e) Optionally you can disable the sudo password and enable auto-login to root. I'd also recommend enabling SSH.
  - (f) Apply the settings.
5. Run `sudo bash installer.sh`
6. When prompted to reboot press Y.

## 1.3 Setting up composite display

1. Run `sudo raspi-config`
2. Select *2 Display Options*
3. Select *D4 Composite*
4. Select *Yes* for "Would you like composite video output to be enabled?"
5. Press *Ok, Finish*
6. Run `sudo reboot`
7. Connect the display to the composite output.
8. Verify that it's working before moving on. At this stage if there's no display output you'll need to SSH into the raspi to figure out what's happening.

## 2 Manual installation

### 2.1 Initial setup

1. Flash an SD card with Rasbian Lite using the raspberry pi imager. Keep in mind that this code should work on other hardware/os versions, but setup will likely differ.
2. Plug in the RPI, connect it to a display via HDMI and give it power via the USB-C port.
3. Connect the RPI to the internet, either via it's ethernet port or WIFI.

### 2.2 Software installs and interface setup

1. Run `sudo apt update` and `sudo apt full-upgrade`
2. Run `sudo apt install git` (i forgot if git is installed by default)
3. Run `git clone https://github.com/Sirusdog/Thermal-camera-project.git`
4. Run `cd Thermal-Camera-Project`
5. Run `pip install -r requirements.txt --break-system-packages`
6. Run `sudo raspi-config`
  - (a) Select *3 Interface Options*
  - (b) Enable *I4 SPI*
  - (c) Enable *I5 I2C*
  - (d) Select *I6 Serial port*
    - i. Select *No* for "Would you like a login shell to be accessible over serial?"
    - ii. Select *Yes* for "Would you like the serial port hardware to be enabled?"
  - (e) Optionally you can disable the sudo password and enable auto-login to root. I'd also recommend enabling SSH.
  - (f) Apply the settings.
7. Run `sudo reboot`

## 2.3 Cythonizing main and setting up auto-run

1. Run `cd Thermal-Camera-Project`
2. Run `python3 initialiser.py`
3. Run `python setup.py build_ext --inplace`
4. Run `sudo nano thermalCamBoot.service`
  - (a) Change the line `ExecStart=/user.bin/python3 /home/sirus/Thermal-camera-project/run.py` so that the path matches the username of your raspi, i.e if you set the pi's username to pi it'll be `ExecStart=/user.bin/python3 /home/pi/Thermal-camera-project/run.py`
  - (b) Press `Ctrl + X` and then `Y` to save the file.
5. Run `sudo cp thermalCamBoot.service /etc/systemd/system/`
6. Run `sudo systemctl daemon-reload`
7. Run `sudo systemctl enable thermalCamBoot` TODO: Check if this starts the program.

Past this point I'd recommend assembling the (using a normal screen and psu) to test if everything software-wise is working.

## 2.4 Setting up composite display

1. Run `sudo raspi-config`
2. Select *2 Display Options*
3. Select *D4 Composite*
4. Select *Yes* for "Would you like composite video output to be enabled?"
5. Press *Ok, Finish*
6. Run `sudo reboot`
7. Connect the display to the composite output.
8. Verify that it's working before moving on. At this stage if there's no display output you'll need to SSH into the raspi to figure out what's happening.

## 3 Parts

### Optical elements

| Item ID | Description                    | Designations |
|---------|--------------------------------|--------------|
| LB1757  | F=30 mm, 25.4 mm biconvex lens | $f_1$        |
| ME1-G01 | 25.4 mm first surface mirror   | $m$          |
| EBS1    | 25.4 mm 50:50 Beamsplitter     | $b$          |
| PL25mm  | 25mm Plossl eyepiece           | $f_2$        |
| LB—     | F=20 mm, 12.4 mm biconvex lens | $f_3$        |

**Note that prices do *not* include shipping/vendor costs.**

### GX16 Connector Pinout

| Pin # | Name        | RPI                       | NVG                                    |
|-------|-------------|---------------------------|----------------------------------------|
| 1     | 5v          | GPIO 5V pins              | Thermal camera 5V in, display 5V in    |
| 2     | GND 1       | GPIO GND pin              | Thermal camera GND, rotary encoder GND |
| 3     | 3.3v        | GPIO 3.3V pins            | Rotary encoder VCC                     |
| 4     | Encoder CLK | GPIO 23                   | Rotary encoder CLK pin                 |
| 5     | Encoder DT  | GPIO 24                   | Rotary encoder DT pin                  |
| 6     | Encoder SW  | GPIO 25                   | Rotary encoder SW pin                  |
| 7     | Cam CVBS    | CVBS-USB board CVBS input | Camera CVBS output                     |
| 8     | Cam RX      | UART0 TX                  | Cam RX pin                             |
| 9     | Cam TX      | UART0 RX                  | Cam TX pin                             |
| 10    | Screen CVBS | RPI composite out         | Screen composite in                    |
| 11    | GND 2       | GPIO GND pin              | Screen GND, brightness controls        |
| 12    | NC          | NC                        | NC                                     |

### Part List - Normal version

| Name                                         | Qty | Cost (€) | Link                              |
|----------------------------------------------|-----|----------|-----------------------------------|
| MX10130 White phosphor autogating NNVT5 tube | 1   | 1385.39  | Alibaba link (likely cheaper now) |
| PVS-7 collimator                             | 1   | 178.46   | Bought off of Ebay                |
| PVS-7 objective lens                         | 1   | 176.14   | Bought off of Ebay                |
| 0.8V-5V to 3V Voltage regulator              | 1   | 6.15     | Aliexpress                        |
| 5mW IR Laser (Line)                          | 1   | 12.46    | Aliexpress                        |
| LB1757 F=30mm, D=25.4mm biconvex lens        | 2   | 28.56    | Thorlabs                          |
| ME-G01 Aluminum Mirror, 3.2 mm Thick         | 2   | 33.33    | Thorlabs                          |
| C&K COMPONENTS A20403RNZQ 4P2S Rotary switch | 1   | 12.1     | Farnell                           |
| Momentary switch                             | 1   | ?        | Lost track                        |
| 25mm Orbinal Plossl Eyepiece                 | 2   | 26.90    | Amazon                            |
| 10mm M3 nuts and bolts                       | 20  | ?        | Hardware store                    |

$P$  = Total bonus

$$\text{Major involvement payout} = \frac{P}{N_{\text{Major investigators}} - 0.5(N_{\text{Minor investigators}})}$$

$$\text{Minor involvement payout} = \frac{P - \text{Major involvement payout} \times N_{\text{Major investigators}}}{N_{\text{Minor investigators}}}$$

### Extended parts list - Thermal version

| Name                               | Qty | Cost (€) | Link                                 |
|------------------------------------|-----|----------|--------------------------------------|
| Mini2 Thermal Camera               | 1   | 174.20   | Ebay (Can be bought from Aliexpress) |
| 0.39in OLED Display (AV tricolour) | 1   | 73.23    | Aliexpress                           |
| KY-040 Rotary Encoders             | 1   | 7.32     | Amazon                               |
| 120 Dupont wires                   | 1   | 12.68    | Amazon                               |
| EBS1 50:50 Beamsplitter            | 1   | 37.27    | Thorlabs                             |
| LB1494 F=12mm, D=9mm biconvex lens | 1   | 26.70    | Thorlabs                             |
| GX16 12 pin connectors male        | 2   | a        | a                                    |
| GX16 12 pin cable female-female    | 1   | a        | a                                    |
| Raspberry Pi 4B 2GB                | 1   | 61.65    | The Pi Hut                           |
| CVBS-USB converter board           | 1   | 38       | Ivcan                                |
| Audio jack to RCA cable            | 1   | 5.02     | The Pi Hut                           |
| Heatsink                           | 1   | 0.93     | The Pi Hut                           |
| 1.25mm Pitch 5-pin 28AWG Cable     | 1   | 1.05     | The Pi Hut                           |

### Extended parts list - GPS, compass & waypoints

| Name                               | Qty | Cost (€) | Link       |
|------------------------------------|-----|----------|------------|
| USB GPS Reciever                   | 1   | 14.94    | The Pi Hut |
| Adafruit BNO085 9-DOF IMU breakout | 1   | 28.02    | The Pi Hut |